

DPCM Predictor Assignment

Communication Systems – Understanding Prediction in Differential PCM

Objective

The purpose of this assignment is to understand how the predictor in a Differential Pulse Code Modulation (DPCM) system works, how the predictor coefficient affects the prediction error, and why good prediction can reduce the dynamic range of the signal that must be quantized.

$$e[n] = x[n] - \hat{x}[n]$$

where $\hat{x}[n]$ is the predicted value of the current sample.

Part A – First-Order Prediction

Consider the discrete-time sequence:

$$x[n] = \{10, 12, 15, 17, 18, 20, 21, 23\}$$

Assume initially that there is no quantization error. Use the predictor:

$$\hat{x}[n] = x[n-1]$$

- A1. Determine the predicted sequence $\hat{x}[n]$.
- A2. Calculate the prediction error $e[n] = x[n] - \hat{x}[n]$.
- A3. Compare the range of $x[n]$ with the range of $e[n]$.
- A4. Explain why quantizing $e[n]$ may require fewer bits than directly quantizing $x[n]$.

Part B – Effect of the Predictor Coefficient

Consider the first-order predictor:

$$\hat{x}[n] = a x[n-1]$$

Investigate $a = 0, 0.5, 1$, and 1.2 . For each value:

1. Determine $\hat{x}[n]$.
2. Determine $e[n]$.
3. Calculate the mean-square prediction error (MSE).

$$MSE = (1/(N-1)) \sum e^2[n]$$

Predictor coefficient, a	Prediction error $e[n]$	Prediction MSE
0		
0.5		
1		
1.2		

Answer:

4. Which predictor gives the smallest MSE?
5. What does $a = 0$ mean physically?

6. Why can an excessively large predictor coefficient produce worse results?
7. Does prediction always reduce the signal amplitude? Explain.

Part C – Finding the Optimum Predictor

For $\hat{x}[n] = a x[n-1]$, show that the coefficient minimizing the squared prediction error is:

$$a_{\text{opt}} = [\sum x[n]x[n-1]] / [\sum x^2[n-1]]$$

Calculate a_{opt} for the sequence in Part A, calculate the corresponding prediction MSE, and compare it with the values obtained in Part B.

Part D – MATLAB Investigation

Consider the sampled signal

$$x[n] = 2 \cos(0.05\pi n) + 0.5 \cos(0.2\pi n), \quad 0 \leq n \leq 199$$

This signal changes relatively slowly from one sample to the next and therefore has significant correlation between adjacent samples.

D1 – Generate and Plot the Signal

Generate $x[n]$ in MATLAB and plot approximately the first 60 samples.

```
n = 0:199;
x = 2*cos(0.05*pi*n) + 0.5*cos(0.2*pi*n);
```

D2 – Simple Predictor

Use $\hat{x}[n] = x[n-1]$. Calculate the predicted signal and prediction error.

Plot $x[n]$ and $\hat{x}[n]$ on the same figure. Use another figure to plot $e[n]$.

Question: From the figures, explain why $e[n]$ is generally smaller than $x[n]$.

D3 – Prediction Gain

$$P_x = (1/N) \sum |x[n]|^2 \quad P_e = (1/(N-1)) \sum |e[n]|^2$$

$$G_p = P_x/P_e \quad G_p(\text{dB}) = 10 \log_{10}(P_x/P_e)$$

Calculate the prediction gain. What does a large prediction gain mean?

Part E – Effect of Predictor Coefficient Using MATLAB

Use $\hat{x}[n] = a x[n-1]$ and test:

$$a = 0, 0.5, 0.8, 1, 1.2$$

For each value, calculate the prediction MSE and plot MSE versus a .

8. Which value of a gives the minimum MSE?
9. Is $a = 1$ necessarily the optimum predictor?
10. What determines the optimum predictor coefficient?
11. What happens when the predictor coefficient is poorly selected?

Part F – Calculate the Optimum Predictor in MATLAB

Calculate a_{opt} using:

```
a_opt = sum(x(2:end).*x(1:end-1)) / sum(x(1:end-1).^2);
```

Then calculate the prediction error using a_{opt} and compare its MSE with the MSE for $a = 1$. Is the difference significant? Explain based on the characteristics of the signal.

Part G – Effect of Signal Frequency

Repeat the experiment using:

$$x_2[n] = 2 \cos(0.6\pi n)$$

Use the simple predictor $\hat{x}_2[n] = x_2[n-1]$ and compare its prediction error with Part D.

12. Which signal is easier to predict using the previous sample?
13. Why?
14. What does this demonstrate about the relationship between signal correlation and DPCM performance?

Part H – Connection to the Complete DPCM System

In a practical DPCM system, the prediction error is quantized before transmission:

$$e[n] \rightarrow \text{Quantizer} \rightarrow e_q[n]$$

$$x_q[n] = \hat{x}[n] + e_q[n]$$

15. Why is the predictor also required at the receiver?
16. Why must transmitter and receiver predictors be identical?
17. Why should the transmitter predictor normally use previously reconstructed samples rather than original samples?
18. What can happen if the transmitter and receiver predictors use different previous samples?
19. Why does DPCM perform particularly well for speech signals?
20. Would DPCM offer the same advantage for completely uncorrelated random samples? Explain.

Bonus – Second-Order Predictor

$$\hat{x}[n] = a_1 x[n-1] + a_2 x[n-2]$$

Investigate whether using two previous samples can reduce the prediction MSE compared with the optimum first-order predictor. Discuss whether increasing predictor order is always beneficial considering prediction accuracy, complexity, coefficient sensitivity, and computational cost.

Required Submission

21. Hand calculations for Parts A–C.
22. MATLAB code.
23. Required plots.
24. Completed MSE comparison table.
25. Short answers to the conceptual questions.
26. A 5–7 line conclusion explaining: What does the predictor actually do in a DPCM system, and why does a good predictor improve DPCM efficiency?